

Independent Application Answer Sheet - Hydrogen Model and Spectrum Evidence

Student: _____

Date: _____

MLA Standard: MLA.CHEM.ATO.03

Directions: Open the PhET Models of the Hydrogen Atom simulation from Moodle. Complete every step in order. This completed PDF is submitted with U2 L6 Notebook Evidence.

Part A - Simulation setup and model comparison

1. Select **Prediction**. Turn on the light source and spectrum display. Compare two model choices. Record only what you observe.

Model used	What the electron does	Spectrum evidence	One limitation or difference
Model 1:			
Model 2:			

2. Which model best represents allowed electron energy levels? Use one observation as evidence.

Part B - Absorption and emission

Observed event	Electron movement	Absorption or emission?	Evidence
Electron moves to a higher level			
Electron moves to a lower level			
A bright spectral line appears			

Independent Application Answer Sheet - Analysis

Student: _____

Date: _____

MLA Standard: MLA.CHEM.ATO.03

Part C - Explain the evidence

3. Explain why the line spectrum contains separate lines instead of every color.

4. Use the staircase analogy to explain quantized energy without saying that electrons literally stand on stairs.

5. Correction check: write one incorrect statement about electron energy, then correct it using **allowed energy level**, **photon**, and **transition**.

Submission check: Save the completed PDF with your name and U2 L6. Upload it to U2 L6 Notebook Evidence Submission.